

Intelligent Smoke Detection System for Fire Safety Systems

1. Abstract

This project focuses on improving IoT-based smoke detection by developing a machine-learning model using the AdaBoost classifier. The dataset contains real environmental sensor readings with noticeable class imbalance and noisy feature distributions. After applying data cleaning, feature selection, class-balancing techniques, and hyperparameter tuning, the AdaBoost model achieved highly reliable performance, with an accuracy of 0.9895, perfect precision (1.0), and a strong F1-score of 0.9928. The results show that AdaBoost effectively distinguishes between fire-related and non-fire smoke conditions while minimizing false alarms. Overall, the findings highlight the potential of ensemble learning methods to enhance the accuracy, stability, and responsiveness of modern smoke-detection systems.

2. INTRODUCTION

Fire detection systems are essential for ensuring safety in residential, commercial, and industrial environments. Despite their importance, many traditional smoke alarms suffer from a critical drawback; they frequently trigger alerts in response to harmless smoke sources such as cooking fumes, cigarettes, or steam. These repeated false alarms not only cause inconvenience but also reduce user trust and may lead to delayed reactions during real fire emergencies.

With the integration of Machine Learning and IoT technologies, there is an opportunity to enhance the

intelligence of fire alarm systems by enabling them to analyze environmental conditions more accurately

and make informed decisions. This project is motivated by the need to develop a more reliable smoke detection mechanism capable of distinguishing between fire-related smoke and non-threatening smoke sources.

3. PROBLEM DESCRIPTION

This project involves building a machine learning model that classifies smoke measurements into two categories: fire-related smoke and non-fire smoke. The system uses real-time sensor data to determine whether the alarm signal should be activated or not, making this a binary classification problem.

The decision is represented by the target column Fire Alarm, where a value of 1 indicates a true fire event and 0 represents non-fire smoke conditions.

The objective of this project is to achieve accurate and consistent alarm prediction by enabling the model to distinguish between different smoke sources, with a primary focus on reducing false alarms (False Positives). By minimizing incorrect alert activation, the system functions as an intelligent decision layer that enhances the reliability and performance of fire detection systems across varied conditions.

4. BACKGROUND STUDIES

1- Andelić et al. (2023) conducted one of the few machine-learning studies using the same Smoke Detection Dataset from Kaggle that our project is based on. Their research focused on developing symbolic classification models that learn mathematical expressions capable of distinguishing

fire conditions from non-fire smoke using sensor-fusion data, including temperature, humidity, TVOC, eCO₂, and particulate matter readings. By applying resampling techniques such as SMOTE and using a symbolic genetic-programming classifier, the study achieved high accuracy and demonstrated the effectiveness of combining multiple environmental sensors for reliable fire detection. The findings show that machine-learning models can capture subtle variations in gas concentrations and particle behavior that traditional smoke detectors often overlook. This supports our project's approach of applying an ensemble method (AdaBoost) to the same type of sensor-based data in order to minimize false alarms and improve early detection performance.

2- Recent studies have highlighted major improvements in smoke-detection systems using machine-learning and sensor data. Sathishkumar et al. (2023) introduced a deep-learning approach for detecting forest fire smoke with high accuracy and stability, showing the importance of using advanced learning techniques to reduce false alarms. Other research explored multi-sensor data fusion, noise-handling methods, and better decision strategies to distinguish between real fire events and harmless smoke in practical environments.

These findings reveal a clear gap in traditional threshold-based smoke detectors, which often fail under noisy or variable conditions. Building on this, our project uses IoT sensor readings and the AdaBoost ensemble method to provide a more reliable and accurate smoke-detection model. By leveraging boosted decision trees, our system aims to minimize false alarms and improve early-fire detection performance.

3- This 2024 review paper discusses modern fire and smoke detection systems that combine Machine Learning (ML) and Internet of Things (IoT) technologies. The study highlights the weaknesses of traditional detection methods and proposes a smart

hybrid system using multiple sensors such as smoke, gas, temperature, LiDAR, ultrasonic, and cameras to enable early and accurate fire detection.

AI models, especially CNNs and YOLO, are used to analyze images and sensor data, reducing false

alarms and improving responsiveness. The paper also introduces an autonomous firefighting robot capable of navigating dangerous areas and assisting in extinguishing fires, reducing risks for human firefighters.

Overall, the research shows that integrating ML and IoT significantly improves detection accuracy, response time, and safety in fire emergencies.

4-The 2023 research paper “Real-time fire smoke detection method combining a self-attention mechanism and radial multi-scale feature connection” presents an intelligent model designed to improve the accuracy and speed of fire-smoke detection. The study explains that traditional detectors often fail with small or unclear smoke patterns, so the authors developed a deep-learning method that enhances feature extraction and focuses on the most important smoke characteristics. The results showed high accuracy and a clear reduction in false alarms. This directly supports our project because both aim to build a smarter and more reliable smoke detection system using advanced machine-learning techniques. Although the research uses image-based features and our project relies on IoT sensor data with the AdaBoost classifier, both approaches share the same goal: improving real-world smoke identification and minimizing incorrect alarms.

4.1 DATA DESCRIPTION

The dataset used in this project is *smoke_detection.csv*, collected from a real IoT-based smoke monitoring system designed to identify early signs of smoke or fire. It originally contains 62,630 samples and 16 columns, including 15 numerical sensor features and 1 binary target variable (Fire Alarm: 0 = No Alarm, 1 = Alarm). The sensor measurements include:

-UTC: a timestamp indicating when each reading was captured.

-Temperature[C]: ambient temperature, which typically rises during smoke or fire events.

-Humidity[%]: moisture level in the air, affecting smoke behavior and particle density.

-TVOC[ppb]: total volatile organic compounds; these chemicals increase sharply when materials burn.

-eCO2[ppm]: equivalent CO₂ concentration, which rises in combustion and smoke-filled environments.

-Raw H2: hydrogen sensor reading; hydrogen levels may increase during electrical or chemical fires.

-Raw Ethanol: ethanol vapor level; certain burning materials release ethanol-like gases.

-Pressure[hPa]: atmospheric pressure, influencing airflow and smoke dispersion.

-PM1.0 and PM2.5: concentrations of fine particulate matter; smoke contains high levels of these tiny particles.

-NC0.5, NC1.0, NC2.5: particle counts for different size ranges, reflecting smoke density and combustion residue.

-CNT: an internal sample counter used by the device; it does not directly relate to smoke but is included in the data.

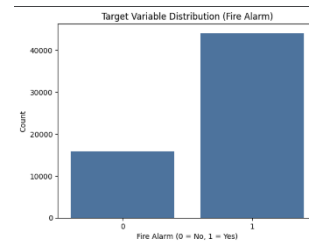
During initial inspection, we discovered an additional exported index column, 'Unnamed: 0', which contained no useful information and was removed. All sensor readings were purely numeric, and the dataset contained no missing values, which simplified preprocessing.

To ensure data quality, several preprocessing steps were applied. First, rows where TVOC = 0 were removed since zero gas concentration is physically invalid and indicate faulty sensor readings. Next, a new engineered feature called Temp_category was created based on the median temperature, classifying each reading as either cold (0) or hot (1) to capture environmental differences. Using the correlation heatmap, we identified that NC2.5, NC1.0, and PM2.5 showed very weak correlation with the Fire Alarm label and were highly redundant with other particle-based features; therefore, they were dropped. After cleaning and refining the dataset, the final version used for modeling consisted of 13 relevant features and 1 target label, providing a clean, consistent, and well-structured dataset suitable for training machine-learning models.

4.2 EXPLORATORY DATA ANALYSIS

The chart shows a clear class imbalance in the Fire Alarm column where class 1(Fire) represented 73.58% of the data while class 0(Not Fire) accounted for 26.42%.

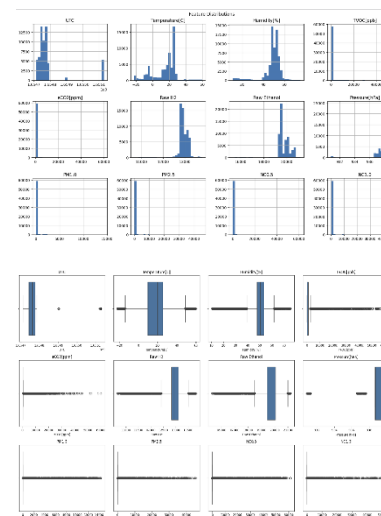
-We plotted histograms to understand the distribution of each numeric feature and detect skewness or unusual



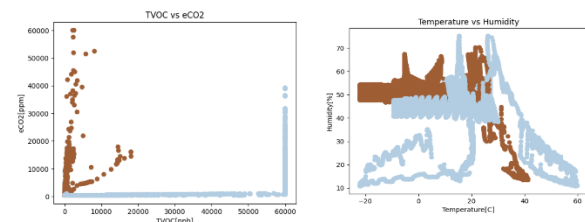
proportion	
Fire Alarm	
1	73.578389
0	26.421611

patterns.

-We used boxplots to identify potential outliers and observe how each feature spreads across its range.



-Scatter plot visually shows how two features behave together, how the two classes (fire/no fire) separate, and reveal trends or clusters.



[illegible]

This project follows a structured process to train a machine learning model that classifies smoke as fire or non-fire using AdaBoost algorithm. The dataset was divided into features and the target column Fire Alarm, then split into training and testing sets using an 80–20 ratio with stratification to preserve class balance. This ensures a fair and consistent evaluation of the model's performance.

After reviewing suitable techniques to handle this issue, SMOTE (Synthetic Minority Over-sampling Technique) was selected as the most effective approach. SMOTE works by generating new synthetic samples for the minority class rather than simply duplicating existing ones. It creates these samples by interpolating between existing minority data points and their nearest neighbors in the feature space, allowing the model to learn more balanced and generalized patterns.

The algorithm used in this project is AdaBoostClassifier, an ensemble learning technique that improves prediction performance by combining multiple weak learners into a single strong model. In this case, a Decision Tree classifier was used as the base estimator.

This learning strategy makes AdaBoost particularly suitable for the smoke detection dataset, which contains overlapping patterns between fire and non-fire conditions. Since some sensor readings share similar ranges across both classes, a model that can adaptively focus on difficult and ambiguous cases is essential. By emphasizing these challenging samples, AdaBoost improves its ability to distinguish subtle differences in smoke characteristics, supporting more reliable prediction of the Fire Alarm target.

Parameter	Value	Description
n_estimators	200	Number of weak learners (trees) in the ensemble. More estimators allow the model to learn more complex patterns.
learning_rate	0.01	controls the contribution of each weak learner to the final ensemble.
estimator_max_depth	1	Limits each decision tree to a depth of 1 (a decision stump), which keeps weak learners simple and suitable for AdaBoost.

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Precision was selected as the scoring metric during tuning to align directly with the project’s objective of minimizing false alarms and improving the reliability of alarm activation decisions. The optimal parameter combination obtained from this process was subsequently used to train the final model

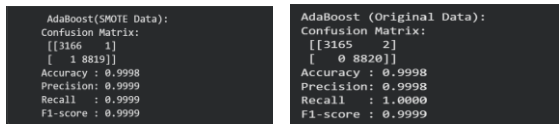
6.Experiments & Results:

To evaluate the performance of the AdaBoost classifier, four configurations were tested to examine the impact of data balancing and hyperparameter tuning:

- 1. Original Data – Before Tuning
- 2. Original Data – After Tuning
- 3. SMOTE Data – Before Tuning
- 4. SMOTE Data – After Tuning

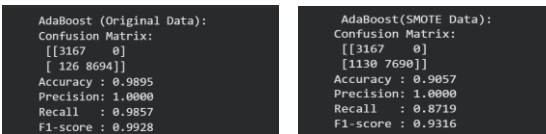
Each configuration was assessed using Accuracy, Precision, Recall, and F1-score. These evaluations helped measure both prediction performance and the model’s generalization ability.

Before Tuning:



• Figure 1: Confusion Matrices — Before Tuning (Original vs SMOTE)

After Tuning:



• Figure 2: Confusion Matrices — After Tuning (Original vs SMOTE)

Performance Comparison

	Model	Accuracy	Precision	Recall	F1-score
0	Original Data - Before Tuning	0.999833	0.999833	0.999833	0.999833
1	SMOTE - Before Tuning	0.999833	0.999833	0.999833	0.999833
2	Original Data - After Tuning	0.989489	1.000000	0.985714	0.992806
3	SMOTE - After Tuning	0.905731	1.000000	0.871882	0.931557

Table 1. Performance comparison across all configurations (original vs SMOTE, before and after tuning).

Best overall performance comes from the original data after tuning, where the model achieved the highest precision and a high F1 score (see Table 1).

We selected this model because our primary objective is to reduce false alarms. Higher precision directly corresponds to fewer false positives, which aligns with the goal of ensuring that the system triggers an alarm only when a true fire event is likely.

Visualization Analysis:

To further evaluate the tuned AdaBoost model, three visualizations were generated: the ROC curve, the colored confusion matrix, and the learning curve (Figure 3).

The confusion matrix shows highly accurate classification, with 3167 correct non-smoke predictions and 8694 correct smoke predictions, and very few misclassifications, indicating strong recall performance.

The ROC curve demonstrates excellent separability between the two classes, achieving an AUC of 0.99, reflecting near-perfect discrimination and confirming that the model can maintain strong predictive performance across different thresholds.

The learning curve shows consistent and stable F1-scores for both the training and validation sets as the training size increases. This indicates that the model generalizes well and does not exhibit overfitting.

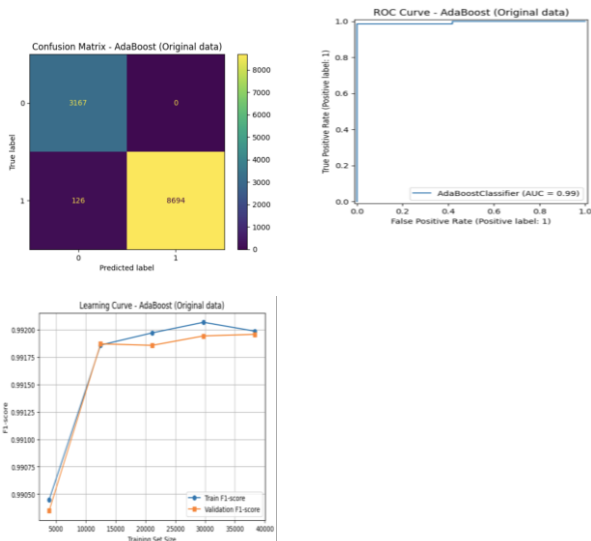


Figure 3. ROC curve, confusion matrix and learning curve for the tuned AdaBoost model on the original dataset.

7.Discussion

Interpretation of Results:

The AdaBoost classifier demonstrated strong predictive performance on the fire-alarm dataset, achieving an accuracy of 0.9895 and an F1-score of 0.9928. The model reached perfect precision (1.0), indicating that it does not generate false alarms. Additionally, its recall of 0.9857 shows that it correctly identifies almost all true fire-alarm events. Overall, the results suggest that AdaBoost successfully captured the complex relationships between the environmental sensor readings and the fire-alarm signal.

Challenges:

Several challenges influenced the modeling process:

- 1. **Class imbalance** : Fire Alarm = 1 represents ~74% of the data, making naïve models appear “good” without actually learning meaningful patterns.
- 2. **Mixed correlations** : Some features had strong positive or negative correlations with the target, while others contributed noise, making feature learning less straightforward.

These challenges highlight the importance of using a robust ensemble model such as AdaBoost.

Comparison to Baseline Models:

Model	Accuracy	Precision	Recall	Strengths	Weaknesses
Dummy Classifier	0.73	Very low	Very low	—	Predicts all samples as “1”; fails on negative class – unusable
Naïve Bayes	0.89	Low	High	Detects most fire events (good recall)	Many false alarms; struggles with noisy sensor data
Logistic Regression	0.997	High	High	Strong and stable baseline; balanced performance	May fail to capture non-linear patterns
AdaBoost (Our Model)	0.9895	Perfect	Strong	No false alarms; very reliable; excellent F1 (0.9928)	Slightly lower accuracy than Logistic Regression

8. Conclusion

The results of this project demonstrate that the AdaBoost classifier is highly effective for IoT-based smoke detection. After proper preprocessing, handling class imbalance, and performing hyperparameter tuning, the model achieved high accuracy, excellent recall, and strong generalization as shown by the learning curve. Comparisons with baseline models confirmed that AdaBoost—particularly after tuning on the original dataset—delivered the best overall performance, proving the suitability of boosted tree-based methods for distinguishing between fire and non-fire smoke conditions.

9. Limitations

Although the AdaBoost model achieved strong performance, there are a few limitations to consider. The model is sensitive to noisy or inconsistent sensor readings, which may affect its stability in real environments. In addition, some features in the dataset show overlapping patterns between fire and non-fire cases, making the classification task more challenging. Finally, the model was trained on a specific IoT dataset, so its performance may vary if applied to different sensors or environments.

10.Future Work

Future improvements may include testing the model on additional sensor types, collecting larger and more diverse datasets, and exploring advanced ensemble or deep-learning techniques to further enhance accuracy and reduce false alarms.

11. References

A. Research Papers

Research Paper1:

N. Anđelić, S. Baressi Šegota, I. Lorencin, and Z. Car, “The development of symbolic expressions for fire detection with symbolic classifier using sensor fusion data,” *Sensors*, vol. 23, no. 1, p. 169, 2023. [Online].Available: <https://doi.org/10.3390/s23010169>

Research Paper 2:

[2] V. E. Sathishkumar, J. Cho, M. Subramanian, and O. S. Naren, "Forest fire and smoke detection using deep learning-based learning without forgetting," *Fire Ecology*, vol. 19, no. 1, 2023. [Online]. Available: <https://link.springer.com/article/10.1186/s42408-022-00165-0>

Research Paper 3:

Medewar, A. G., Sawarkar, A. D., & Kshirsagar, U. V. (2024, September 26). *A review on fire and smoke detection with intelligent control for enhanced safety using machine learning (ML) and Internet of Things (IoT)*. *Cureus J Comput Sci*, 1: es44389-024-00359-4. <https://doi.org/10.7759/s44389-024-00359-4>

Research Paper 4:

C. Jin, A. Zheng, Z. Wu, and C. Tong, "Real-time fire smoke detection method combining a self-attention mechanism and radial multi-scale feature connection," *Sensors*, vol. 23, no. 6, 2023. [Online]. Available: <https://doi.org/10.3390/s23063358>

B. Dataset

[1] DeepContractor, *Smoke Detection Dataset*, Kaggle, 2020. [Online]. Available: <https://www.kaggle.com/datasets/deepcontractor/smoke-detection-dataset>

C. Algorithm Reference

[1] Y. Freund and R. E. Schapire, *A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting*, *Journal of Computer and System Sciences*, vol. 55, no. 1, pp. 119–139, 1997.